Biomedical Signal Processing and Control 123 (2026) 110410

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

XAAI-ledger: An explainable CNN-transformer-based multi-modal deep learning framework for early detection of melanoma and non-melanoma skin cancers using dermoscopic and clinical data

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ARTICLE INFO

Keywords:
Skin cancer
Artificial intelligence
Explainable AI (XAI)
Deep learning
Convolutional neural network (CNN)
Transformer encoder (TE)

ABSTRACT

Over the past few decades, skin cancer—particularly melanoma—has become a significant global health problem due to its rising incidence and challenging diagnosis. It is believed that improving patient outcomes requires both accuracy and early detection. This paper presents XAAI-Ledger, an explainable and hybrid Convolutional Neural Network (CNN) Transformer-enabled multi-modal deep learning framework that combines clinical metadata and dermoscopic images to categories skin cancers, including melanoma and non-melanoma. In addition to using CNN to extract spatial features, the Transformer Encoder (TE) in this proposed solution aids in collecting semantic linkages and long-range dependencies across medical record modalities. To improve interpretability and decision accuracy, visual and non-visual elements are combined using an attention-based fusion method. Explainable AI (XAI) mechanisms, such as SHapley Additive exPlanations (SHAP) and Gradient-Weighted Class (Grad-CAM) Activation Mapping, are utilized to foster therapeutic trust. These mechanisms not only make it possible to visualize model thinking but also offer efficacy and dependability. Simulation findings on multiple benchmark datasets, particularly ISIC 2019, show a 98.71% classification accuracy, a 97.45% sensitivity, and a 98.22% specificity. When compared to alternative cutting-edge techniques, it takes into account the superior metrics that this proposed model achieves. In summary, by offering a transparent, resilient, and high-performance solution that is appropriate for dynamic clinical environments, the proposed XAAI-Ledger advances AI-enabled dermatological diagnostic procedures.

1. Introduction

Melanoma is one of the most aggressive and potentially deadly types of skin cancer, which is one of the most common cancers worldwide. According to the worldwide health report, greater exposure to genetic predisposition, UV radiation, and delayed diagnosis are the main causes of the current indication of both melanoma and non-melanoma skin cancers [1–3]. These days, dermatologists deal with the difficult issue of the aforementioned element. Thus, early, precise, efficient, and trustworthy detection continues to be a dubious perspective for improving

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https://doi.org/10.1016/j.bspc.2026.110410
Received 26 September 2025; Received in revised form 29 November 2025; Accepted 20 April 2026
Available online 9 May 2026

the survival ratio and reducing invasive interventions. However, the current diagnostic cycle mainly depends on clinical expertise and subjective interpretation of dermoscopic images, both of which can vary significantly among practitioners [3,4].

Current developments in artificial intelligence (AI), particularly deep learning (DL), have shown great promise for improving medical diagnostics, notably in the field of dermatology [5,6]. Convolutional neural networks, or CNNs, are essential for demonstrating efficacy in identifying spatial hierarchies in medical images, as shown in Fig. 1. Transformer-based learning encoders (TE) are particularly good at representing semantic relations and long-range dependencies [6,7]. However, current AI methods for classifying skin cancer are frequently restricted to unimodal data, which is important because they prioritize image analysis over structured clinical metadata such as patient history, age, gender, and lesion location.

Early detection is essential for research into developing a viable treatment for skin cancer, primarily melanoma and non-melanoma. However, it has drawbacks, such as the need for faster and more efficient diagnostic hierarchy, the difficulty of present diagnostic techniques, and the dependence on manual examination by dermatologists [8,9], as shown in Fig. 1. When examining skin disorders, there are several variations that could lead to a mistake and further complicate early detection attempts.

Nevertheless, the list of challenges is highlighted and discussed as follows [8–11] (as also illustrated in Fig. 1):

- **Distinguishing Benign and Malignant Lesions**: When assessing skin lesions, rashes and moles may seem harmless, but they may be early indicators of skin cancer.
- **Lesion Variation and Characteristics**: It is worth noting that the melanomas and related skin cancers can present with a wide range of appearances that limit the development of a single, universally applicable detection method.
- **Interpretability of Artificial Intelligence Existing Models and Limitations:** In the identification of melanoma skin cancer, artificial intelligence is demonstrating promise; current models may have trouble balancing precision and recall and dealing with data imbalances such overfitting.
- **Computational Costs and Complexity:** Time-consuming hyperparameter tuning and the need for substantial computational resources, including computational processing, are two issues that arise during the training and implementation of sophisticated AI models.
- **Patient Awareness and Behavior:** Regular self-examinations are essential for early detection since they raise awareness of their significance.

In order to address such type of challenges and limitations, this paper proposes XAAI-Ledger, an explainable, multi-modal DL framework that collaborates with both dermoscopic images and clinical metadata for the classification of melanoma and non-melanoma cancers. This proposed solution leveraging the strengths of CNN for extraction of local features, along with transformers for global contextual learning, as shown in Fig. 1. Due to this, the proposed model collects a large and more comprehensive representation of the data regarding diagnostics. An attention-enabled fusion strategy is developed to efficient, effective, and reliable integration of visual and non-visual features, which improves both model interpretability and predictive performance.

The increasing need for transparent artificial intelligence in the healthcare environment is another recent response received from the experts. While designing XAAI-Ledger, it is evident to incorporate XAI features in terms of Grad-CAM and SHAP to provide visual insights into the model's process of effective and reliable decision-making, as shown in Fig. 1. Due to this, the proposed solution receives improved clinical trust but also ensures regulatory compliance for potential real-time applications. On the other end, this proposed solution is validated with the use of ISIC 2019 skin cancer dataset, along with the multi-benchmarking in order to achieve a classification accuracy up to 98.71, which is considering a remarkable mark in this field, Along with that, this solution maintain the sensitivity up to 97.45% and specificity up to 98.22% compared to other state-of-the-art methos. Such enhancements and better result fluctuations demonstrate the model's reliability and effectiveness, as well as its superiority over current methods. Overall, XAAI-Ledger offers an interpretable, high-performance, and novel solution for AI-driven skin cancer diagnostics, particularly for melanoma and non-melanoma.

1.1. Motivation of the study

The rising burden of skin cancer classification, especially the early diagnosis of melanoma cancer, underscores the urgent need for an accurate and reliable diagnostic solution, where the role of experts and their intervention is more focused. However, a dermoscopic viewpoint is now a common non-invasive application possibility for assessing skin lesions. The efficiency of the applicational environment must be improved, as it is still heavily reliant on dermatologists' knowledge, which introduces subjectivity and inter-observer variability. Whereas the low-resource setting creates a lack in training professionals while further exacerbating delayed and incorrect diagnoses, which leads to preventable outcomes and enhances the cost of treatment.

On the other hand, DL techniques, especially the use of CNN, have illustrated one of the solutions that automatically classify skin lesions from dermoscopic images. Nevertheless, such a classified technique is often limited in scope, due to focusing solely on image data while ignoring large contextual information embedded in clinical metadata, like lesion location, age of the patients, medical history, etc. This type of unimodal solution restricts the capabilities of the design in terms of potential diagnostics, particularly in ambiguous and early-stage cases, where the context of clinical records management is set to be pivotal, as highlighted in Fig. 2.

In addition, one of the natures of DL models is a 'black box' that raises concerns in clinical practice, where trust, integrity, transparency,

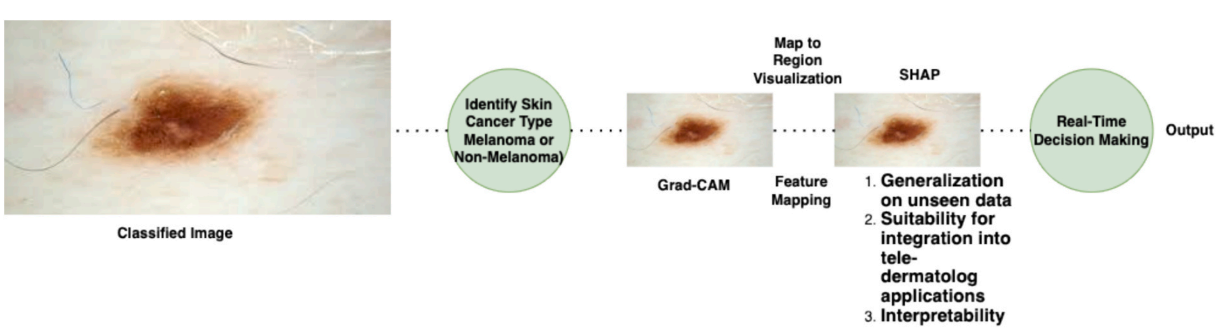

Fig. 1. The Current Individual Classifier Mechanism and Related Execution.

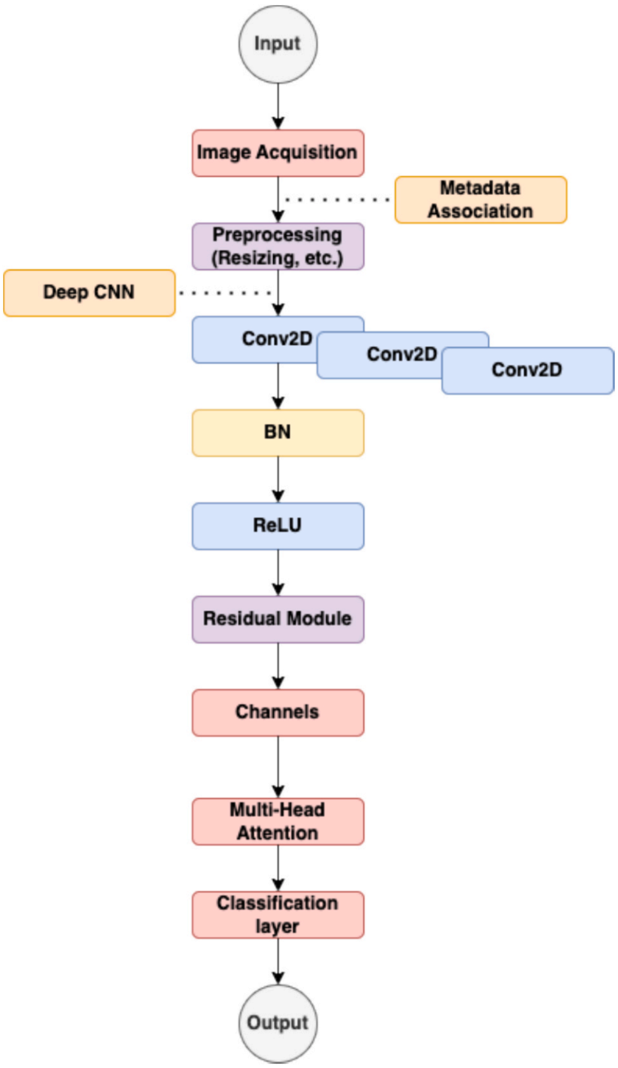

Fig. 2. The Steps of the Proposed Solution.

and explainability are critical for medical decision-making. Healthcare experts, along with dermatologists, are more likely to adopt AI-driven applications when the model's decision can be clearly justified and understood. However, such a type of challenge motivates the implementation of XAAI-Ledger, a novel explainable multimodal DL framework that collaborates clinical metadata and dermoscopic imagery using a hybrid CNN-Transformer hierarchy, as shown in Fig. 2. This design mainly focuses on both visual and non-visual, like contextual information via attention mechanisms that support interpretability using XAI. Nevertheless, the proposed framework allows for creating a model that focuses on bridging the gap between algorithmic performance and clinical applicability. By evaluating all such prospects, we must say that this proposed research is designed to achieve the goal of not only improving the diagnostic accuracy but also ensuring the AI application achieves trust, transparency, and provenance, and aligns with the needs of modern dermatological practices, as shown in Fig. 2.

The major research questions are addressed as follows:

1. How does a multi-modal DL framework effectively solve the critical problem of dermoscopic images and clinical data integration in order to enhance the skin cancer classification?
2. What crucial factors are provided by the hybrid CNN Transformer solution in order to improve feature extraction and contextual understanding of skin lesions?
3. How does an attention-enabled fusion technique improve the overall performance in terms of interpretability?
4. How can XAI association, especially the implication of Grad-CAM and SHAP, incorporate the designed model capability in terms of visual explanation and transparency for diagnostic decisions?

1.2. *Objectives and Contributions of this study*

The major objective of this research is highlighted as follows:

- To design a hybrid CNN-enabled Transformer-based solution that effectively extracts special and contextual features from dermoscopic images and data, as shown in Fig. 2.
- To develop an attention-enabled fusion mechanism for collaborating image features, along with structured clinical data.
- To address XAI integration in order to provide a reliable and interpretable outcome.
- To evaluate the performance of the proposed solution using multi-benchmark datasets and compare with other state-of-the-art methods.
- To contribute to clinical adoptability and scalable AI diagnostic applications.

However, the major contribution of this paper is mentioned as follows:

- This paper presents a novel CNN-Transformer-enabled multi-modal DL framework that integrates dermoscopic images with structured clinical metadata for accurate and reliable skin cancer classification, as shown in Fig. 2.
- This paper introduces a customized attention-based feature fusion mechanism that effectively integrates contextual and visual modalities and allows for designing a model as per the requirements of a dermatologist.
- This paper implements SHAP and Grad-CAM for interpreting designed model predictions, along with improving clinical trust.
- The proposed parameters and their hierarchical customization of this XAAI-Ledger are highlighted as follows:
- Input Image (I) belongs to (heigh (h) * weight (w) * channels (c))
- Input Meta Records (M) belongs to (Clinical Data (cd) * Number of features (d) | {age, sex, location of lesion})
- Image Features (Fimg) = CNN (I) belongs to {the number of extracted image patches | Feature Token (n) * Dimension of feature vector (f)
- *Multi-Head Self Attention [Fcon, w', b] = ReLU (Fcon * w' * b * t * d)*
- This paper presents a design that is mainly customized for clinical deployment and a user-trust centric solution, which makes it suitable for adoption in a real-time dermatology environment.

The following is the structure of this paper's reminder: The current system for classifying melanoma and non-melanoma skin cancers by dermatologists is covered in Section 2. Papers on the list are highlighted and discussed. In addition, the resources and methodology used to create this proposal are described in Section 3. The simulations, findings, and associated discussion were provided in Section 4. Section 5, however, evaluates and discusses the list of open research questions and implications concerns. Finally, Section 6's conclusion and future direction mark the end of this paper.

2. Related work

The existing application of AI in dermatological diagnostics has rapidly grown towards an advanced level, driven by the increasing availability of large-scale dermoscopic image datasets and enhanced in DL models [12,13]. Various papers [12–16] have explored automatic skin cancer evaluations in terms of classification that uses predefined

CNNs due to their strong capability in visual data extraction and investigation purposes. On the other hand, most developing countries have adopted traditional mechanisms to analyze skin cancer and related diagnostics. At the same time, the developed countries are more focused on the process of integration of both, like classical methods with DL association. By evaluating the traditional method, it is worth noting that the dermatological examination requires the proper dermoscopic hierarchy and biopsies. While DL perform different, especially CNNs in terms of offering automatic detection and classification. Throughout this, this model analyzes vast image datasets, because of this, the model achieves high accuracy in identifying cancerous lesion.

2.1. CNN-enabled models for skin cancer detection

In paper [17] presented the differences with the use of deep CNNs and CNNs model for identifying, evaluating, and classifying skin lesions, especially melanoma classification, where both the models achieving dermatologist-level accuracy large-scale image datasets. Since then, frameworks like ResNet, Inception, and DenseNet have been widely adopted for non-melanoma detection [18,19]. Nevertheless, such models are typically limited to unimodal inputs, focusing solely on dermoscopic images without considering complementary clinical data. In order to address the challenges of image-only models, various current papers [19–21] have begun to explore multi-modal learning, which not only integrating clinical metadata but maintaining automation, especially the association of patient age, sex, location of lesion and their severity impacts related data. On the other side, the paper [22] demonstrated that the successful integration of metadata with image features improved classification performance. Along with that, several models employ naïve concatenation solutions, mainly for data fusion purposes, which often fail to capture intricate inter-modal relationships.

With the use of Transformers in Natural Language Processing (NLP), their use in vision tasks has gained momentum [23,24]. Vision Transformers (ViT) and hybrid CNN-Transformer models have been shown to outperform CNNs in collecting global dependencies in medical images [23–25]. The paper [26] proposed a Transformer-enabled dermatology model that achieved superior performance but lacked explainability and clinical metadata collaboration. Explainability is a critical factor in clinical AI adoption. Models and mechanisms like Grad-CAM, SHAP, and local interpretable model-agnostic explanations [27,28] have been widely used to interpret CNN outcomes. Nevertheless, the aforementioned techniques are limitedly extended to multi-modeling and related framework association, especially those involving Transformer-enabling CNN. The lack of transparency in high-performing models remains a key barrier to clinical deployment.

The key research gaps that are evaluated, highlighted, and discussed are as follows [28–30]:

- Most of the current models rely on unimodal inputs
- Requires giant databases
- Limited diagnostic accuracy
- Lack sophisticated multi-modal fusion hierarchy
- Reduces metadata to auxiliary roles
- Explainability in multi-modal AI applications
- Lack of standardization in multi-modal reasoning

3. METHODS and materials

The datasets utilized to train and evaluate this proposed model are first covered in this section. It is worth noting that a dataset has over 1000 datapoints with 3 subsets with 10 attributes of picture data points together with a metadata identifier, it is considered to be one of the benchmarks (such as ISIC). The following is a mention of the dataset list:

- Dataset ISIC proposed by Kaggle (Link: https://www.kaggle.com/datasets/nodoubttome/skin-cancer9-classesisic).
- Skin Cancer MNIST: HAM10000 proposed by MINIST (Link: https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000).
- Melanoma Datasets proposed by CDAS-Cancer.Gov (Link: https://cdas.cancer.gov/datasets/plco/11/).
- Cancer datasets and tissue pathways proposed by RCPath.org (Link: https://www.rcpath.org/profession/guidelines/cancer-datasets-and-tissue-pathways.html).

However, the size of the datasets for training and testing is described as follows:

- Training = 75% of the mentioned data.
- Testing = 25%.

3.1. Notations, problem description, and formulation

This section discusses the mathematical implications of the proposed XAAI-Ledger framework, which mainly integrates dermoscopic image features and structured clinical metadata using a hybrid CNN with a transfer mechanism, along with the attention-based fusion strategy. Such integrational prospects help to design a novel customized model as per the problem of this research evaluation. The successful mathematical formulation of each component of the proposed model is mentioned as follows:

Initially, the input constraints of dermoscopic image and the meta records are defined as follows (also described in Table 1 and illustrated in Fig. 3):

$$Ie(h{*}w{*}c) \tag{1}$$

$$Me(cd{*}Numberoffeatures(d)|\{age, sex, locationoflesion\}) \tag{2}$$

Along with that, the backbone of the designed CNN with the integration of EfficientNet extract hierarchical features from the input dermoscopic images in accordance with (as shown in Fig. 3):

$$\begin{aligned} &ImageFeatures(Fimg) = CNN(I)belongsto \\ &\{thenumberofextractedimagepatches|FeatureToken(n) \\ &{*}Dimensionoffeaturevector(f)\end{aligned} \tag{3}$$

The M is passed through a layer which is fully connected to project it or help in displaying images into the same embedding space as follows (as shown in Fig. 3):

$$\begin{aligned} MetaFeatures(Fmeta) &= \sigma(wightmetric(w') + bias(b))|R; where\sigma \\ &= \text{Activation function (ReLU)}\end{aligned} \tag{4}$$

Table 1
Notations.

Notations	Explanation
I	Input dermoscopic images
h	Height
w	Weight
c	Channels
M	Meta Records
cd	Clinical data
d	Number of features
CNN	Convolutional Neural Network
Fimg	Image features
b	Bias
w'	Weight metrix
Fmeta	Meta Features
R	Predefined evaluation criteria
f	Concatenated features
ff	Fused features
t	Time to extracted feature

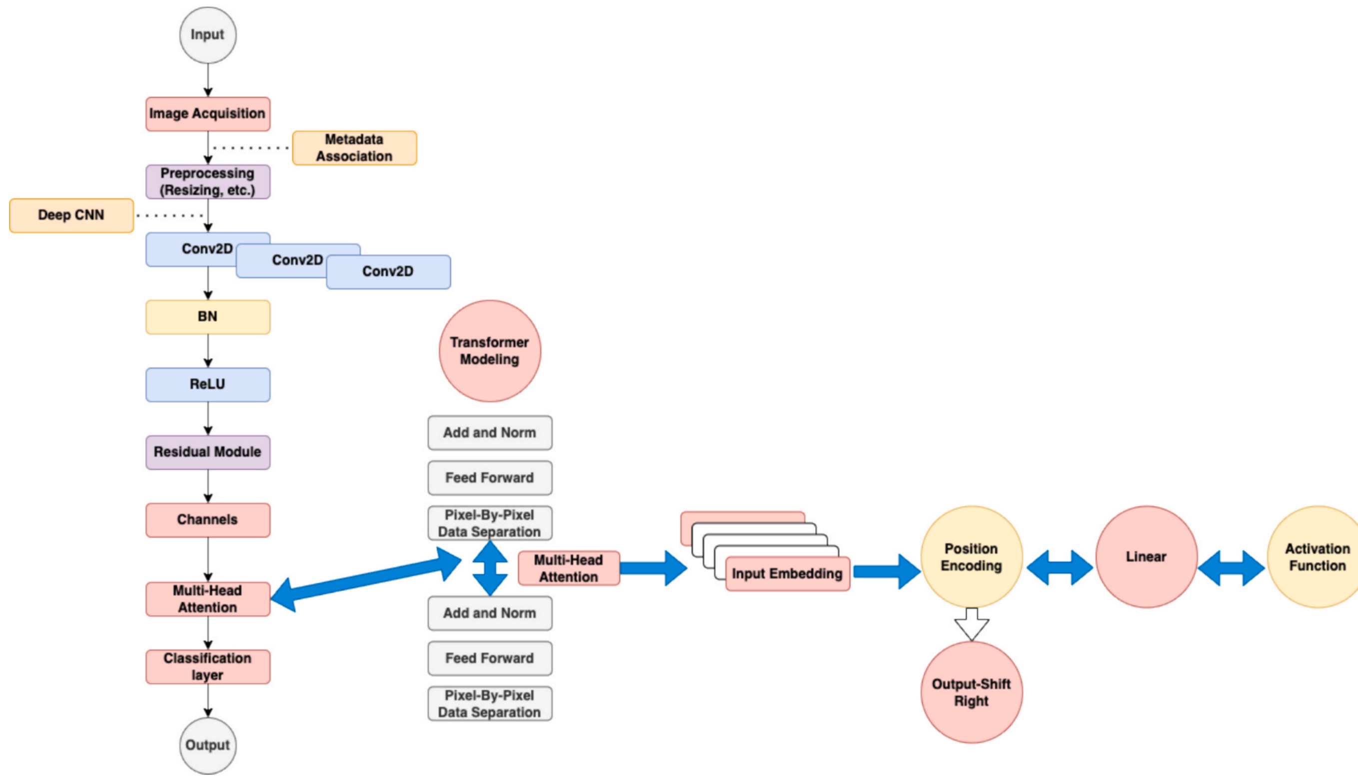

Fig. 3. The Process of the Proposed Melanoma and Non-Melanoma Skin Cancer Early Detection.

However, we integrate the image and metadata features as follows:

$$FeatureIntegration(Fcon) = \{(\mathrm{Fimag} + \mathrm{Fmeta}) | \mathrm{R}\ (\mathrm{n} + 1)^{*}\ \mathrm{f}\} \quad (5)$$

Such integrated features are fed to the encoder of Transformer to collect global context and inter-modality relationships (as shown in Fig. 3):

$$FusedFeatures(ff) = TransformerEncoder(Fcon) \quad (6)$$

Throughput this, each encoded transformer layer consists of multi-head self-attention hierarchy as follows (as illustrated in Fig. 3s):

$$Multi - HeadSelfAttention[Fcon, w', b] = ReLU(Fcon^{*}w'^{*}b^{*}t^{*}d) \quad (7)$$

The following feedforward neural mechanism processes the fused representation after it has been aggregated and combined with the global average pooling with CLS token (as illustrated in Fig. 3):

$$Probabilityforrepresentingmelanoma, non - melanomaclasses(y) = ReLU(ff + w' + b) \quad (8)$$

Nevertheless, the categorical cross-entropy loss is used as follows:

$$Cross - EntropyLoss(L) = \sum\{y'^{*}log(y)\}; wherey'istheground - truthlabel \quad (9)$$

On the other end, computes 'C' gradients of the output class "y' * C", activation maps 'AM' in the last CNN layer is defined as follows:

$$AM = \sum \partial\{\mathrm{y}'^{*}\ \mathrm{C}\} \quad (10)$$

$$Grad - CAM = ReLU(\sum\ \mathrm{AM}) \quad (11)$$

Along with that, computing contribution score for individual function of metadata by estimating Shapley value over input values. Whereas it offers a globule and local explanation for the proposed model behavior.

4. Proposed model

The proposed solution's operation is shown in Fig. 4, where the initialization step is based on obtaining dermoscopic pictures and related clinical metadata, including age, sex, and lesion site, from the multi-benchmark datasets of ISIC 2019 using MINIST HAM1000. As stated in Table 2 and shown in Fig. 4, the model first performs pre-processing, which includes handling missing data through imputation, resizing and normalizing dermoscopic images, encoding categorical metadata, and normalizing numerical metadata. In the next step, the proposed model maintains visual features extraction with the use of CNN. In this whole process, a deep CNN with EfficientNet is used to extract hierarchical spatial features from the images, which is a dermoscopic visual of melanoma or non-melanoma. However, the output of this step is managing a feature map representing the visual characteristics of the skin lesion and related location.

On the other side, the process of clinical metadata embedding is designed, where the proposed solution passes the structured clinical metadata via a fully connected dense layer to transform it into a feature vector of the same dimension as the CNN output. Due to this, we receive a unique dense feature representation of the clinical data. Fig. 5 present the interconnectivity of the proposed multi-modal feature fusion through transformer, where the model is capable for concatenating the image features and metadata features into a unified sequence. It mainly passes the integrated features through a Transformer encoder, which captures (i) global dependencies, (ii) inter-modality interactions, and (iii) context-aware relationships via multi-head self-attention.

In the next step, the model applies a global average pooling, along with the use of special classification token from the transformer output, like melanoma and non-melanoma. This classification layer feed the pooled representation into a fully connected ReLU classifier to identify and predict the class. However, we train the complete model using the 'Adam' optimizer, along with the categorical cross-entropy loss. By achieving this process, this model applies regulation technique by adopting batch normalization in order to prevent overfitting. Due to

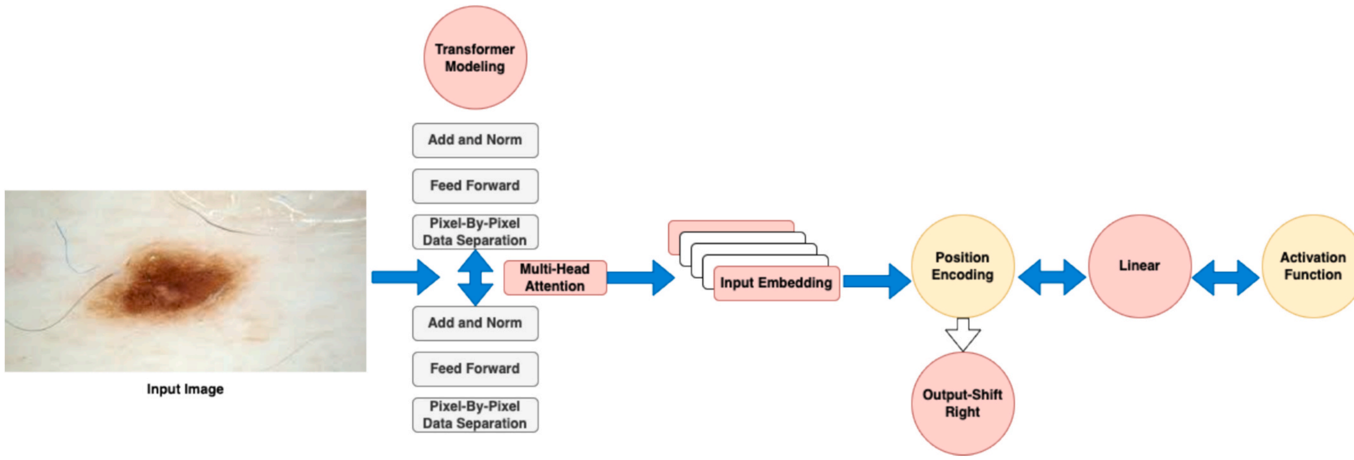

Fig. 4. The Proposed CNN-enabled Transformer-Based Solution.

Table 2
Code representation.

```
I ← Dermoscopic picture
M ← Clinical metadata vector
y' ← Ground truth label are the inputs.
y ← Classification predicted (Melanoma or Non-Melanoma Skin Cancer) ← Interpretation in tabular and visual form (Grad-CAM + SHAP)
M ← Normalise and encode metadata
M ← Normalise and resize image I
F_img → CNN(I)
F_meta ← ExpandDims(F_meta)
FullyConnected(M) ← Equilibrium shape for fusing
Integrate (F_img, F_meta) → F_con
TransformerEncoder(F_concat) F_fused
y ← ReLU(y_logits)
y_logits ← FullyConnected(GlobalFeatures) GlobalFeatures
GlobalAveragePooling(F_fused)
CrossEntropy (y, y')
GradCAM(CNN, I, y).
Values of SHAP ← SHAP(F_meta, y)
```

this, we can monitor the performance of data validation and related metrics, such classification accuracy, sensitivity, and specificity, along with area under curve.

At the end of this, we apply Grad-CAM to the output of the CNN features maps to visualize attention regions in the images of dermoscopic. Whereas the SHAP is utilized to identify the contribution of individual clinical features (like generalization on unseen data, suitability for integration into tele-dermatology tools, and interpretability) in real-time decision-making process, as shown in Fig. 5 and Fig. 6. Because of this advancement, this model achieve interpretability via visual and tabular explanations.

5. Simulation results

This paper discussed the software and hardware requirements to ensure the test's success before initialting the simulation process. The following is a list of the needs that are met:

- System Information:
  - o A 13th-generation vPro CPU with a clock speed of 4.2 GHz
  - o 16 GB of RAM
  - o 1 TB SSD that connects to a 4 TB HDD are used to set up the simulation
  - o A dedicated graphic card is also attached, and all of these alternatives are evaluated
  - o Windows 11 platform
  - o Network bandwidth of at least 40 Mbps to 1 Gbps
- Software Requirement
  - o Python 3.9.11
  - o TensorFlow
  - o Keras
  - o NumPy
  - o Matplotlib

However, the simulation results of the proposed XAAI-Ledger-enabled solution are displayed in Fig. 7, where we determine the exact type of skin cancer for early diagnosis by testing for melanoma skin problems. The original image is shown in Fig. 7, the grey scale convergence for more precise skin lesion classification is shown in Figs. 8 and 9, the impact of skin cancer and the analysis of skin regions is shown in Fig. 10, the type of cancer classification—that is, whether it is melanoma or not, is shown in Fig. 10, and the name of the skin cancer and the level it affects is shown in Fig. 11, as Output with Description. However, the outcome of the metadata evaluation for real-time decision making is shown in Fig. 12. The melanoma skin cancer assessment test is represented by these simulations.

Fig. 13 shows the simulation results of the proposed XAAI-Ledger-enabled solution, in which we test for non-melanoma skin issues to identify the precise type of skin cancer for early diagnosis. Fig. 13 displays the original image; Figs. 14 and 15 display the greyscale convergence for more accurate classification of skin lesions; Fig. 16 displays the impact of skin cancer and the analysis of skin regions; Fig. 17 displays the type of cancer, i.e., whether it is non-melanoma; and Fig. 18 displays the name of the skin cancer and the level it affects as Output with Description. However, Fig. 19 displays the results of the metadata evaluation for real-time decision making. These simulations depict the

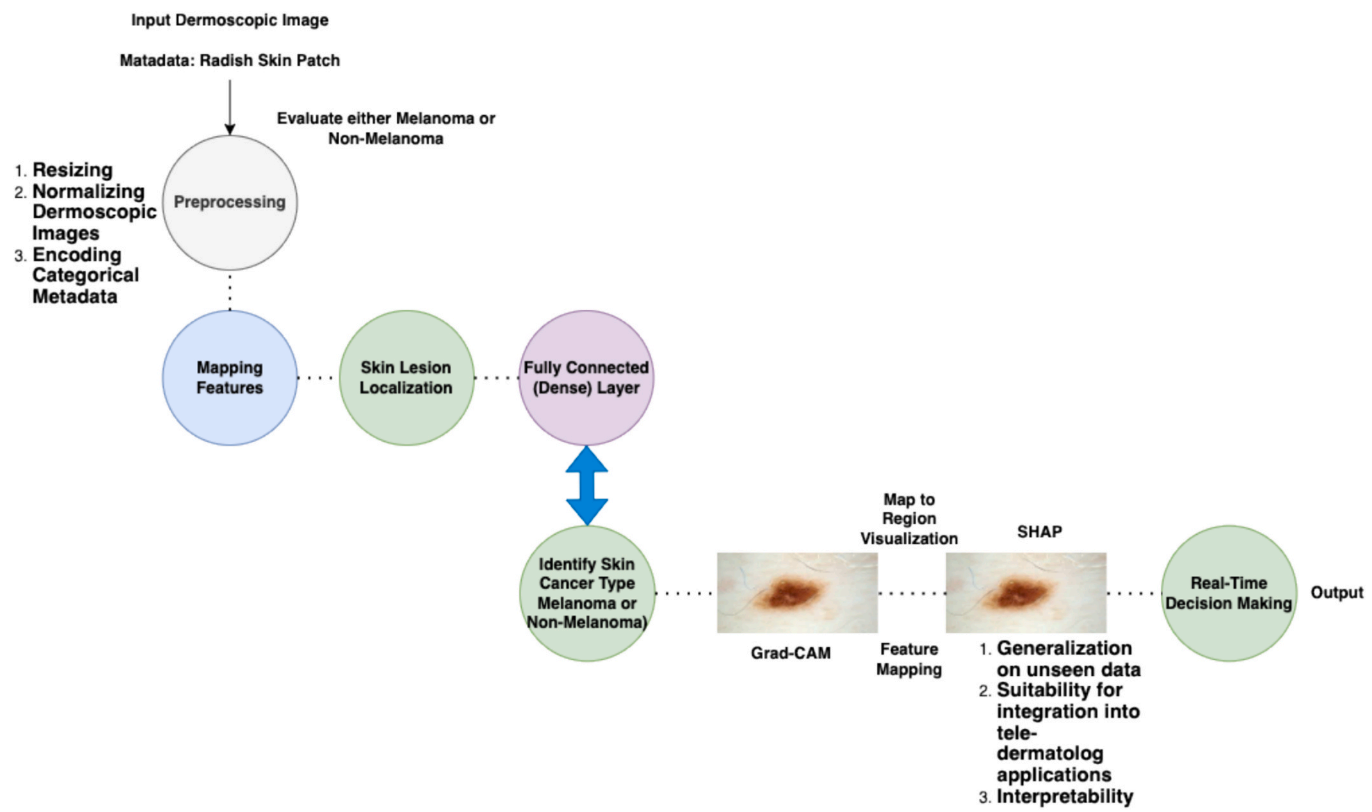

Fig. 5. The Working Execution of Grad-CAM and SHAP.

non-melanoma skin cancer evaluation exam.

For melanoma classification purpose, a similar procedure is used to assess the working cycle of the proposed deep CNN with Transformer Encoder, as shown in Figs. 19 and 20. Through preprocessing, interpretability, localization, explainability, classification, and parameter fluctuation, we start the data load, dermoscopic picture capture and analysis, and rate. The outcome is the best of all; over 600 images are loaded and taken from databases (ISCI 2019 and MINIST HAM1000), and over 1400 and 2000 channels are analyzed to ensure precise classification.

A similar procedure is used to assess the working cycle of the proposed deep CNN with Transformer Encoder for non-melanoma classification, as shown in Figs. 21 and 22. Through preprocessing, interpretability, localization, explainability, classification, and parameter fluctuation, we start the data load, dermoscopic picture collection and analysis, and rate. The outcome is the best of all; over 600 dermoscopic images are categorized along with their meta records created and stored in the databases (ISCI 2019 and MINIST HAM1000), and over 3500 channels are evaluated to ensure accurate categorization in terms of improved results due to metadata association. (See Fig. 23.).

The simulation results on several benchmark datasets (as illustrated in Fig. 19), especially ISIC 2019 and MINIST HAM1000, provide a 98.71% classification accuracy, a 97.45% sensitivity, and a 98.22% specificity after more than 50 iterations of the proposed solution. It considers the better metrics that this proposed customizable model attains in comparison to other state-of-the-art methods [31–33].

The Fig. 24 and Fig. 25 show the pace of image loading and analysis, including interpretability, sensitivity, explainability, localization, and classification of melanoma and non-melanoma skin cancer, as well as parameter variation during the detection process. The outcome is the greatest of all; over 2000 and 14,000 channels' results were received, and more than 600 photos were taken from databases. These findings are regarded as enhanced results since metadata is successfully integrated with dermoscopic images, ensuring precise categorization, which provide a good result compared to other state of the art methods [34–36]. However, based on the findings displayed in Figs. 24 and 25, we must conclude that this XAAI-Ledger approach is a strong contender for real-time industrial adaptation when compared to other cutting-edge techniques [37–39].

In order to evaluate the reliability and effectiveness of the proposed XAAI-Ledger framework and their customize design of the tunned model of CNN-Transformer, a comparative analysis is conducted against different state-of-the-art methods, along with multi-model deep learning techniques using multi-datasets, especially ISIC 2019 skin cancer dataset. The criterion of comparative study is highlighted as follows (as discussed in Table 3, Table 4, and Table 5):

- ResNet50
- DenseNet121
- MobileNetV2
- Multi-Modal CNN
- Vision Transformer
- Hybrid CNN-Transformer

6. Implementation challenges

There are several significant challenges associated with using XAAI-Ledger, particularly when it comes to practicing simulation management. The problem of data availability and quality, however, is one of the main restrictions. Melanoma instances are under-represented compared to benign and non-melanoma lesions in public databases such as ISIC, which frequently have imbalances in terms of data mismanagement and associated variation [40–42]. Such erratic data can bias training and testing towards the majority classes, which not only lowers the sensitivity of detecting malignant lesions but also has no effect on early diagnosis, which is crucial for dermatologists and survival. In addition, dermoscopic images vary widely in resolution, lighting, and acquisition devices, which mainly creates domain shift, which most

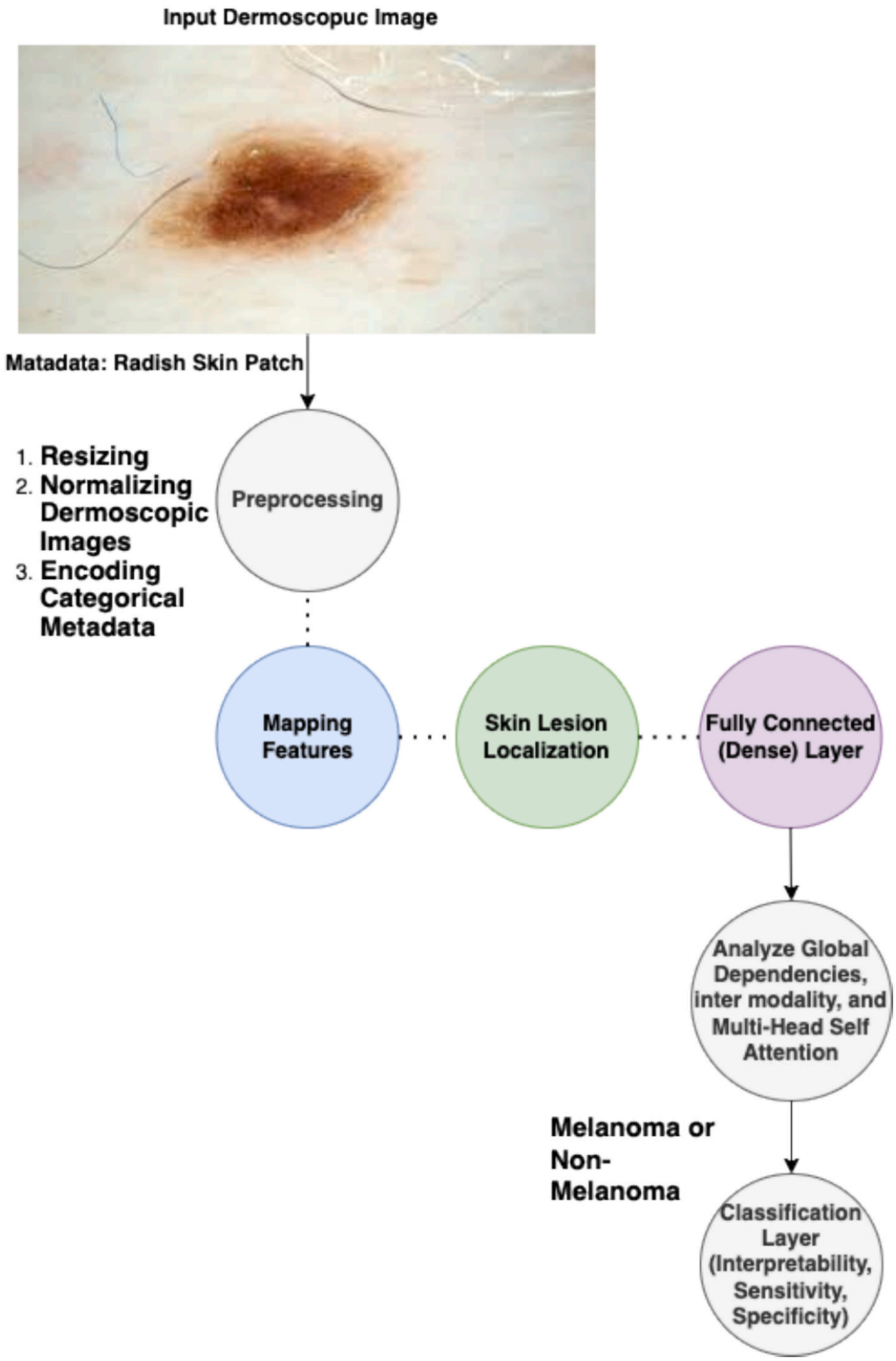

Fig. 6. The Working Objective of Classifier Layer and the Role of Grad-CAM and SHAP.

probably affects the generalization hierarchy. The clinical metadata accompanying such datasets is mostly inconsistent and incomplete [41,43]. Due to this, it necessitates robust preprocessing interconnectivity, along with the architectural design to handle missing information.

On the other side, complexity of multi-modal feature fusion is one of the highlighted challenging problems. Although CNNs are useful for efficiently extracting localized spatial information, it is still challenging to integrate these representations with organized clinical data [44–46]. Although TE-based fusion modules are effective at modelling cross-model relationships, they come with a higher computational cost and are more likely to overfit when data is scarce. One of the crucial issues is striking a balance between the goals of preserving each mode. Additionally, it is evident that it may place an excessive amount of focus on metadata or image attributes, both of which indicate poor performance. Finally, include explainability elements that are critical for clinical adoption, such as Grad-CAM and SHAP. It complicates delays in dynamic environments that demand resource constraints by adding additional computational burden.

7. Open research problems

In order to fully realize the promise of XAAI-Ledger, a number of outstanding research topics need to be resolved. The creation of durable and adaptable fusion processes that continue to function even when faced with varied, sparse, and noisy input modalities is one of the most

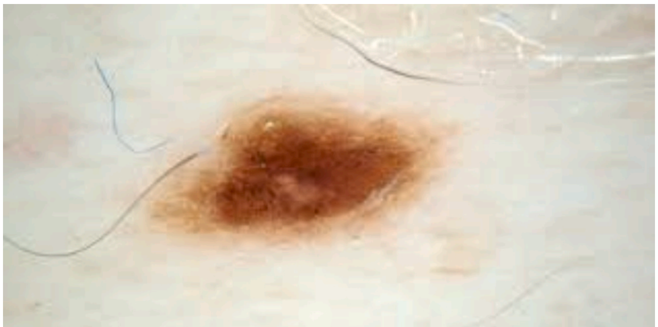

Fig. 7. The Original Image.

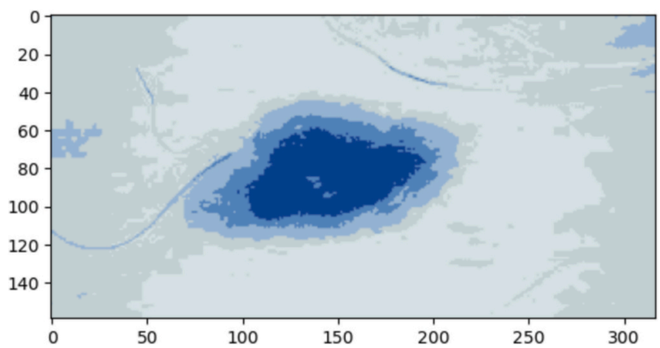

Fig. 8. Gray Scale Convergence for Skin Lesion Analysis (1).

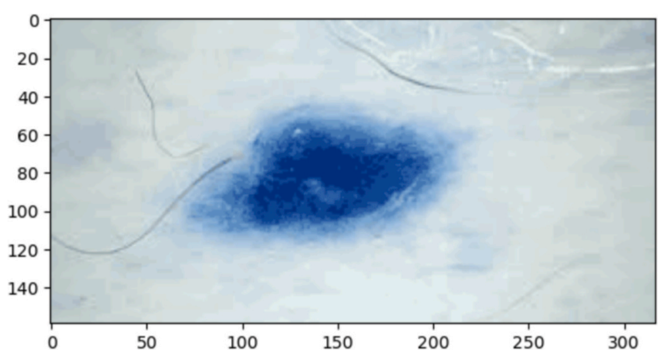

Fig. 9. Gray Scale Convergence for Skin Lesion Analysis (2).

urgent outstanding problems [47–49]. However, the majority of current research focusses on domain adaptation and TL to improve model resilience in the face of changes in imaging technology, clinical procedures, and population demographics [50–52]. It guarantees explainability while maintaining accuracy, which is still a drawback.

Furthermore, privacy, ethical considerations, and prejudice reduction are essential. AI-enabled diagnostic applications, on the other hand, run the danger of inheriting device-specific and demographic biases seen in training data, which could result in disparate diagnostic performance across patient groups.

Fig. 10. Skin Region Evaluated.

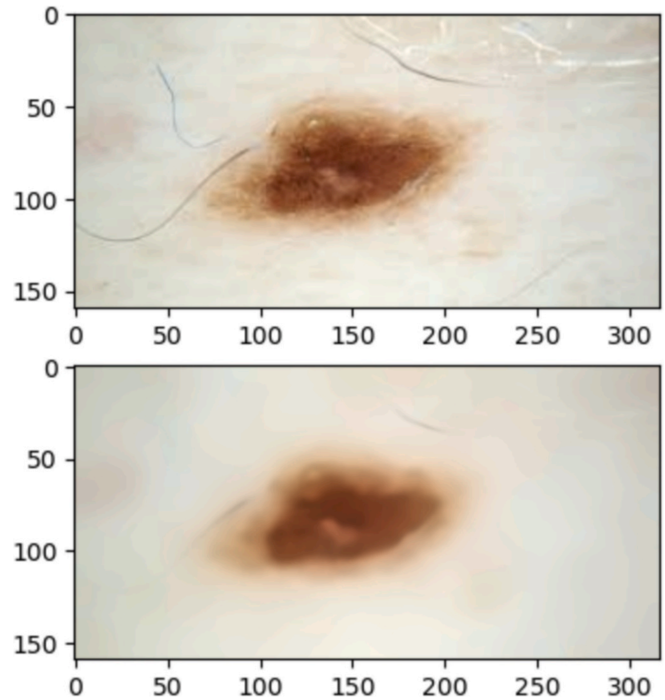

Fig. 11. Melanoma skin Cancer Examined.

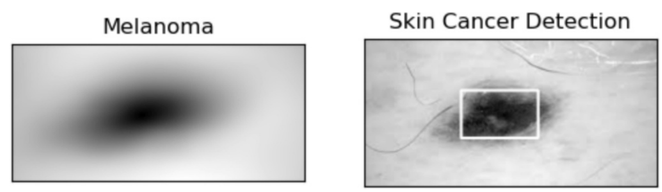

Fig. 12. Metadata for Real-Time Decision Making.

8. Conclusion

The research examined the shortcomings and contemporary patterns dermatologists have noted in the diagnosis and severity assessment of skin cancer. This study highlights and discusses a number of difficult problems, including classification accuracy, sensitivity, specificity, and explainability. On the other hand, this paper introduced XAAI-Ledger, an explainable Transformer-based multi-modal DL framework enabled

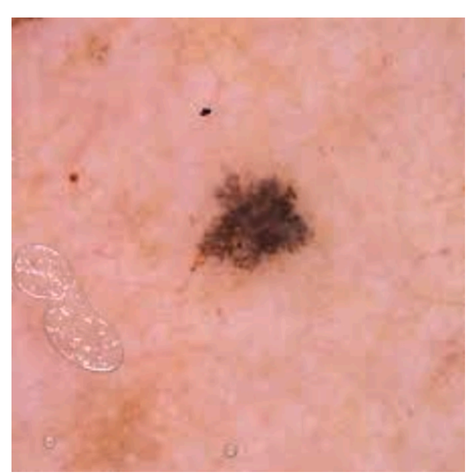

Fig. 13. Original Image.

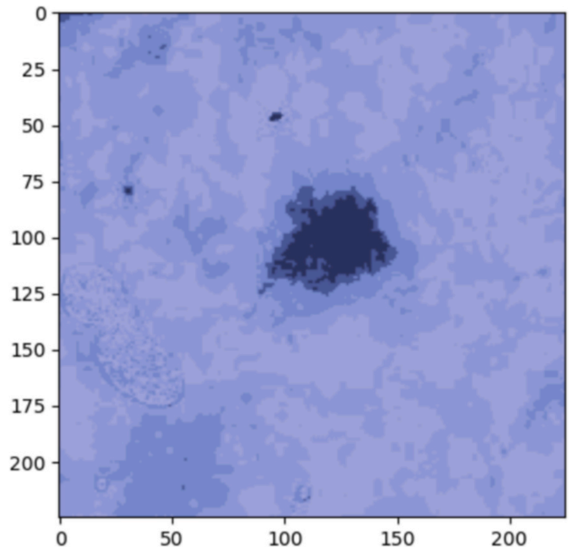

Fig. 14. Gray Scale Convergence for Skin Lesion Analysis (1).

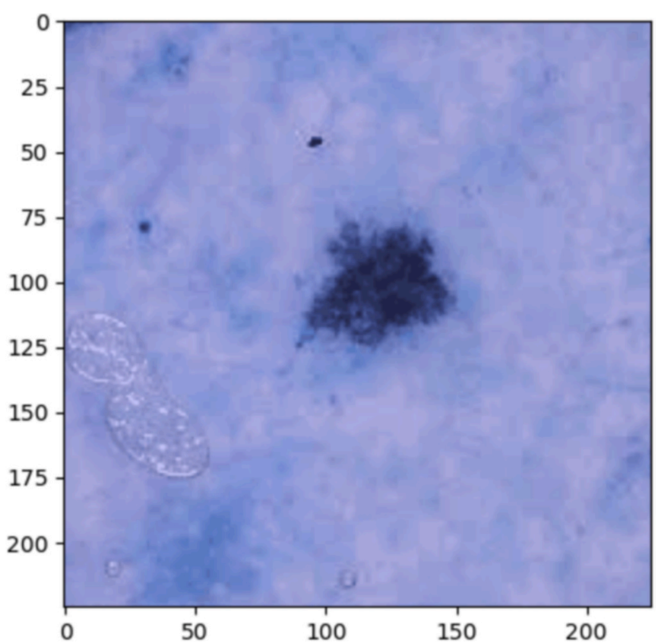

Fig. 15. Gray Scale Convergence for Skin Lesion Analysis (2).

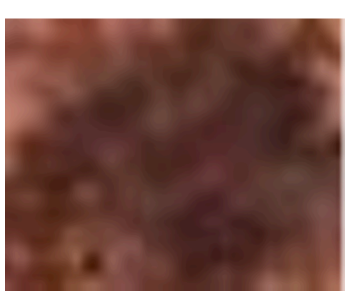

Fig. 16. Skin Region Evaluated.

by CNN that effectively enables a customised model design that integrates dermoscopic images and clinical metadata for the early detection and classification of skin cancer, both melanoma and non-

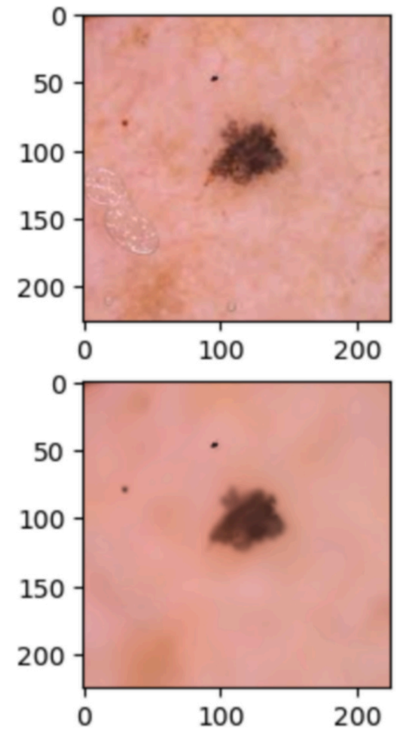

Fig. 17. Non-Melanoma skin Cancer Examined.

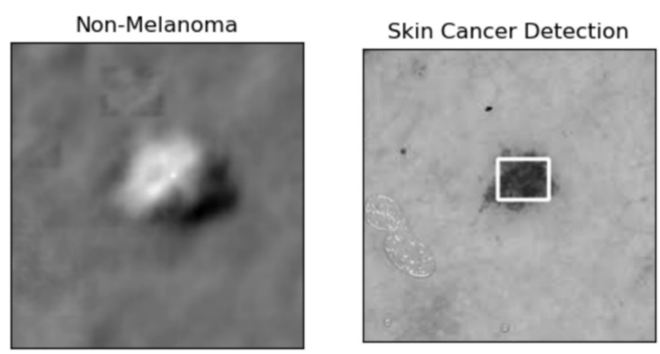

Fig. 18. Metadata for Real-Time Decision Making.

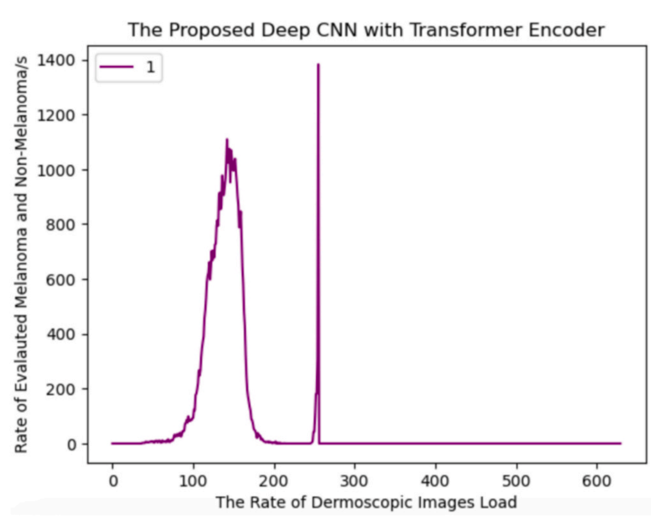

Fig. 19. For Melanoma Analysis-The Rate of Dermoscopic Images Loading (Test 1).

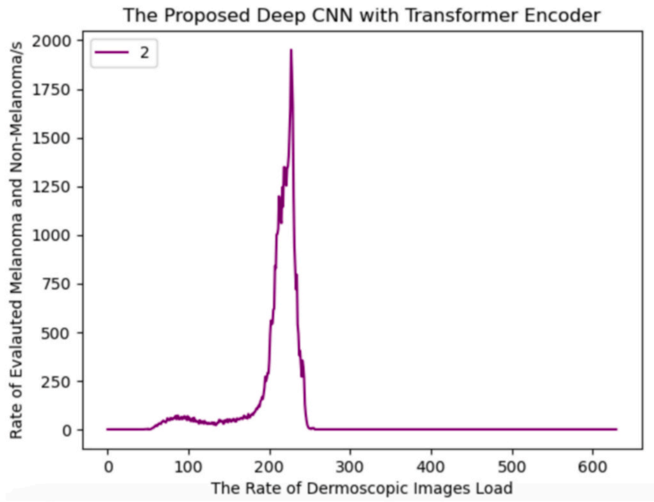

Fig. 20. For Melanoma Analysis-The Rate of Dermoscopic Images Loading (Test 2).

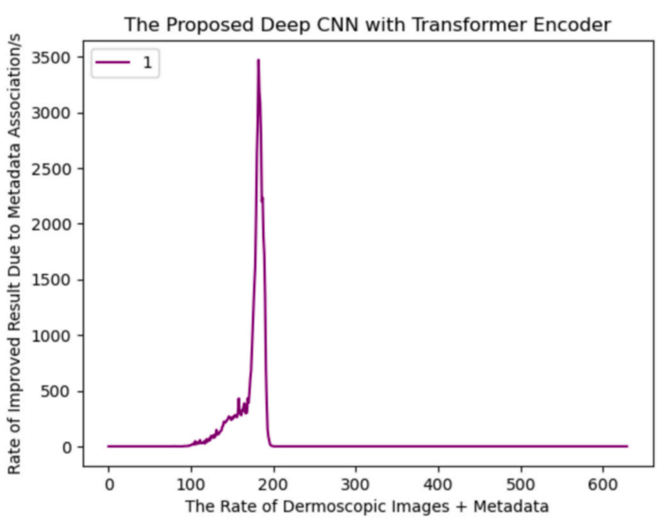

Fig. 21. For Non-Melanoma Analysis-The Rate of Dermoscopic Images + Metadata Evaluated for Improved Result Analysis (Test 1).

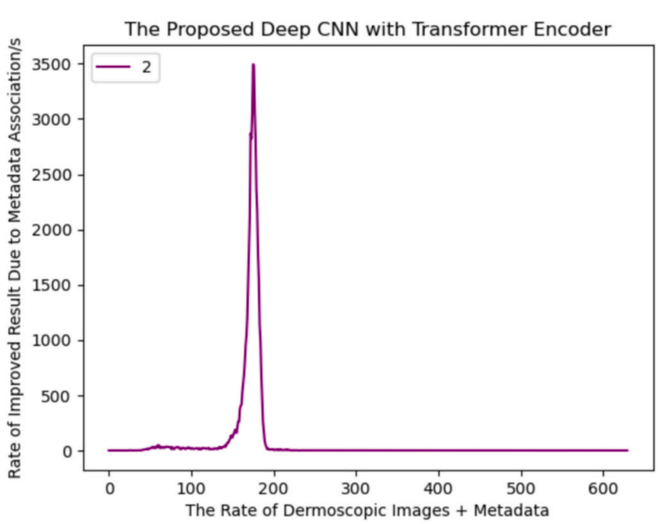

Fig. 22. For Non-Melanoma Analysis-The Rate of Dermoscopic Images + Metadata Evaluated for Improved Result Analysis (Test 2).

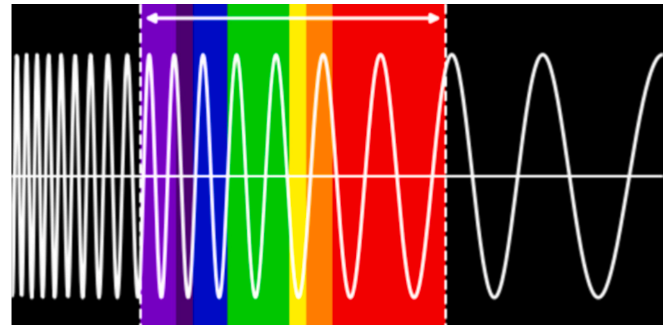

Fig. 23. The Load of Data and Their Size Increment (Black = 1% to Red = 100%).

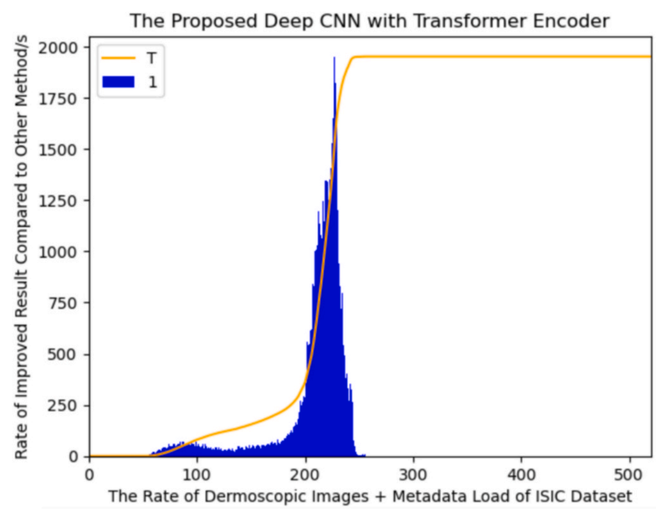

Fig. 24. The Rate of Improved Results Received and Compared (1).

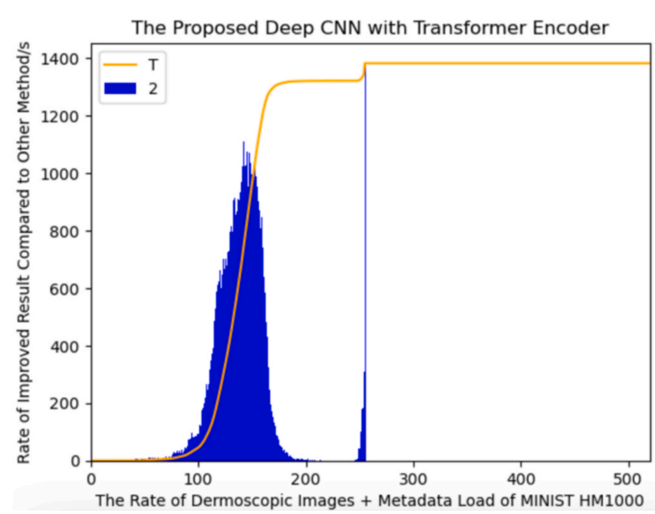

Fig. 25. The Rate of Improved Results Received and Compared (2).

melanoma. In addition to using CNN to extract geographical information, the TE in this proposed solution assists in gathering semantic connections and long-range relationships across medical record modalities. An attention-based fusion method is used to merge visual and non-visual elements to increase interpretability and decision accuracy. To promote therapeutic trust, XAI techniques like Grad-CAM activation mapping and SHAP are used. These mechanisms provide effectiveness

Table 3
Comparison with number of methods/models.

List of methods/models	Explainable
ResNet50	It is a method of deep CNN that most probably used for image-based skin cancer evaluation.
DenseNet121	It is one of the efficient architectures of CNN with dense connections, often used in dermoscopic image analysis.
MobileNetV2	A lightweight CNN must be installed with the aid of MobileNetV2 in order to examine image-only performance and assessment while implementing the model for skin cancer evaluation, particularly in edge deployment.
Multi-Modal CNN	The classical CNN with metadata concatenation for fusion is the main purpose.
Vision Transformer	ViT, a Transformer model implemented to image-only classification without metadata.
Hybrid CNN-Transformer	It mainly integrates CNN and Transformer for improved feature learning, especially melanoma and non-melanoma from clinical images.

Table 4
Comparative analysis test (1).

Constraints of evaluation	Classical CNN	CNN with multi-model	Hybrid CNN with transformer-based deep learning
Sensitivity	No	Yes	Yes
Classification Accuracy	Yes	Yes	Yes
Specificity	Yes	Yes	Yes
Area Under Curve	No	No	Yes
Support of Explainability	N/A	N/A	Yes

Table 5
Comparative analysis test (2).

Constraints of evaluation	Classical deep learning	Deep convolutional neural networks	Hybrid CNN with transformer-based deep learning
Sensitivity	N/A	N/A	Yes
Classification Accuracy	Yes	Yes	Yes
Specificity	No	Yes	Yes
Area Under Curve	No	No	Yes
Support of Explainability	No	No	Yes

and reliability in addition to enabling the visualization of model thinking. Using several benchmark datasets, especially ISIC 2019, simulation results demonstrate a 98.71% classification accuracy, a 97.45% sensitivity, and a 98.22% specificity.

8.1. Future direction

However, depending on the XAAI-Ledger's success, the following deficiencies can be investigated in subsequent studies:

- Ethics and Privacy drawback
- Explainability enhancement
- Robustness and generalization
- Dynamic clinical development and exchange data policy
- Integration mechanism for clinical workflows and related management
- Expanded multi-modal data sources, organization, and optimization

10. Research involving Humans and/or Animals

Not applicable.

11. Consent for publication

No human subjects are harmed in this research, and we confirmed that all data shared with the participants.

There is no objection from the participants. All the participants/consents are agreed on the published version of this research.

Funding Statement

This work was supported by Universiti Teknikal Malaysia Melaka (UTeM) and the Ongoing Research Funding program, (ORF-2026-395), King Saud University, Riyadh, Saudi Arabia.

CRediT authorship contribution statement

Ikram Ul Haq: Writing – original draft, Software, Methodology. **Humayra Atia Joarder:** Investigation, Methodology, Writing – review & editing, Conceptualization. **Abdullah Ayub Khan:** Investigation, Methodology, Writing – review & editing, Conceptualization. **Xuyang Shi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jamil Abedalrahim Jamil Alsayaydeh:** Writing – review & editing, Methodology, Visualization. **Mohd Faizal Bin Yusof:** Writing – review & editing, Methodology, Visualization. **Masrullizam Bin Mat Ibrahim:** Writing – review & editing, Methodology, Visualization. **Hannoud Almoghrabi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ahmad Ali AlZubi:** Writing – review & editing, Methodology, Visualization.

9. Ethics approval and consent to participate

The committee of the University confirmed that all experimental protocols were approved by the organization.

It is confirmed that the experiments follow the criteria of ethics approval and consent to participates.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Annexure A: List of ACRONYMS

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Acronym	Explanation
CNN	Convolutional Neural Networks
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
Grad-CAM	Gradient-Weighted Class
DL	Deep Learning
TE	Transformer Encoder
SHAP	SHapley Additive exPlanations
ViT	Vision Transformers
NLP	Natural Language Processing

Data availability

Data will be made available on request.

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